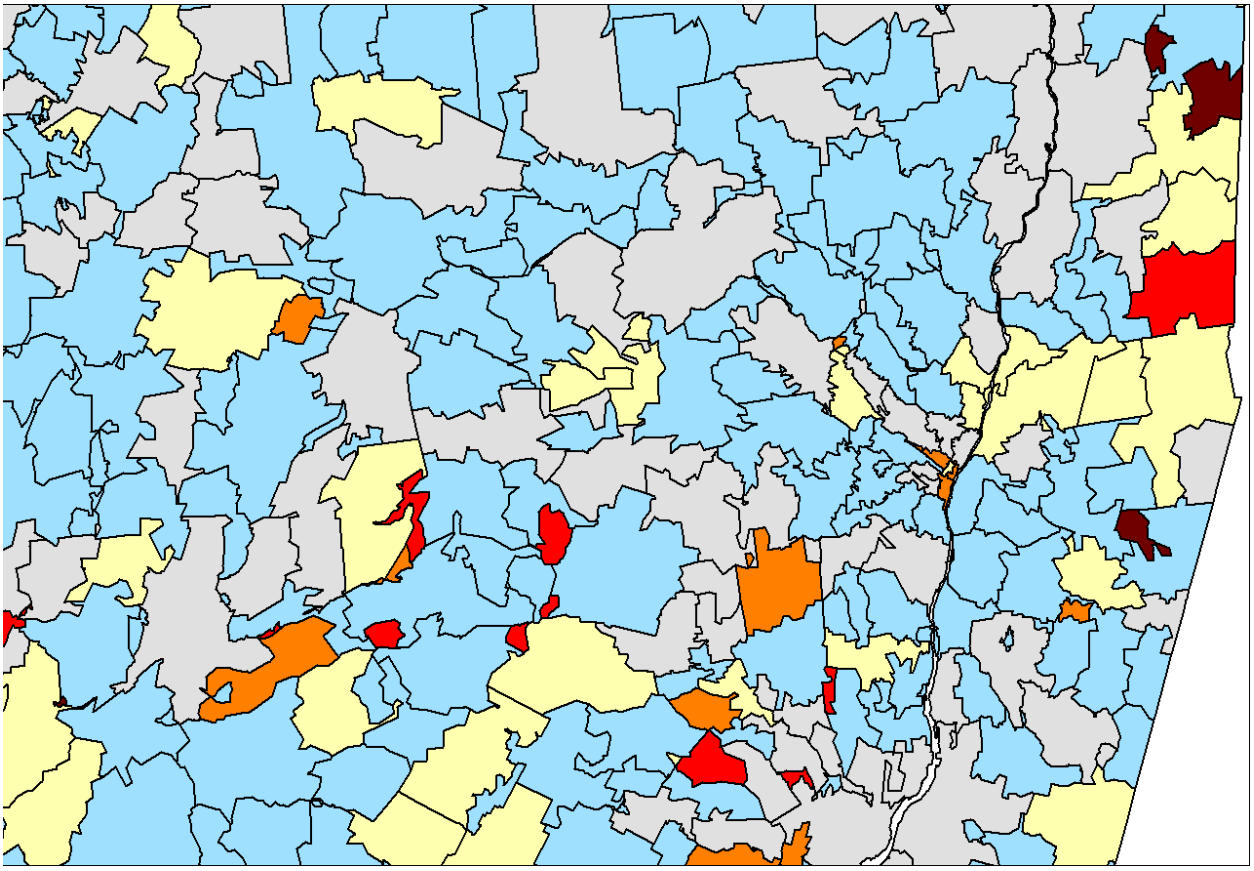
HANDS-ON TRAINING

Geographic Aggregation Tool (GAT)

Workshop handout for· gatpkg version 2.2.0 · November 2026



Presenters

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About the Geographic Aggregation Tool

> **Questions?** Contact Gwen LaSelva (gwen.laselva@health.ny.gov) or Wanhsiang Hsu (wanhsiang.hsu@health.ny.gov).

Why aggregate?

Health outcome maps with fine geographic resolution can be misleading because small numbers cause random fluctuations in disease rates. In some cases, these maps can also inadvertently disclose confidential data. To overcome these limitations, the New York State Department of Health (NYSDOH) Bureau of Environmental and Occupational Epidemiology (BEOE) developed the Geographic Aggregation Tool (GAT). GAT joins neighboring geographic areas until a user-defined population and/or number of cases is reached, so that maps can be published at the finest geographic resolution practicable.

You can restrict merging so that areas are not unnecessarily combined across larger regions. For example, you may want to merge census tracts into larger areas without letting those areas cross county boundaries, or prevent merging across physical features such as mountain ranges or large water bodies.

What you need

GAT has been tested in R 4.5 and R4.6 and may run under other R versions. No R programming experience is required. You need an ESRI shapefile that contains:

- the geographic boundaries of the areas you want to merge;
- a **character variable** that uniquely identifies each area; and
- a **numeric variable** to merge by, such as the number of disease cases or the population.

What GAT produces

GAT outputs a shapefile of the aggregated regions with their associated data, which can be opened in GIS programs such as ArcGIS or MapInfo. In the output:

- Most numeric fields are **summed** across the merged areas.
- Fields named `Latitude`, `Longitude`, `x`, `y`, `Lat`, `lon`, or `long` are **averaged** across the merged areas.
- Character variables (other than the identifier) take the value of the area **last merged** into the region.
- Two new variables, `GATx` and `GATy`, hold the geographic centroids calculated by GAT.
- Areas that were not merged keep their original ID; aggregated areas receive an arbitrary unique ID beginning with "GAT".

A Settings file and a PDF are also produced. Optionally, a KML file may also be produced.

Disclaimer, credits, and funding

Disclaimer. GAT is provided as is. We welcome suggestions for improvement and bug reports.

Credits. If you use GAT, we would appreciate a citation. Converting GAT to an R package would not have been possible without the support of the Centers for Disease Control and Prevention (CDC) National Environmental Public Health Tracking Program and NYSDOH Bureau of Environmental and Occupational Epidemiology (BEOE) staff and interns. Major contributors include Abigail Stamm (NYSDOH BEOE), Gwen (Babcock) LaSelva (NYSDOH BEOE), and Thomas Talbot. Additional contributors include Neil Muscatiello (NYSDOH BEOE), Arjita Rai, Catherine Wargo (NYSDOH BEOE), Douglas Done (NYSDOH BEOE), Maxime Fokoua Dongmo, Wanhsiang Hsu (NYSDOH BEOE), Isaac Finkell, and Marine Nalbandyan. Beta testers include Angela Werner (CDC), Aya Abdelfattah, and Anne Pysnik (NYSDOH BEOE).

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Training Setup

In this training you will use GAT to aggregate small areas into larger ones until a user-defined number of population is reached. Spatially aggregating data in this way reduces random fluctuations in disease rates.

Software and files

Required software	R 4.0 or later
Required R packages	BARD, boot, class, classInt, coda, deldir, e1071, gpclib, lattice, maptools, MASS, Matrix, nlme, RColorBrewer, rgdal, sp, spam, spdep, sfdep, svDialogs, svMisc, tcltk
Training shapefiles	`hftown.shp`: Exercise 1 (installed with gatpkg in `gatpkg/inst/extdata`) `NS_GAT_DEMO.shp`: Exercise 2 (download from GitHub; see Exercise 2)
Optional software	GIS software (e.g., MapInfo, ArcMap, ArcGIS) to view the output shapefiles

How GAT walks you through a run

GAT is a point-and-click program. After you launch it from R, a series of numbered dialog windows ask for each setting. Every dialog has **< Back**, **Next >**, **Cancel GAT**, and **Help** buttons. Before anything is processed, Step 11 shows all your choices so you can go back and change any of them.

Exercise 1: Merging Towns by Population-Weighted Distance

In this exercise you will merge towns so that each final area has 6,000 to 15,000 residents, without crossing county lines, and map population density.

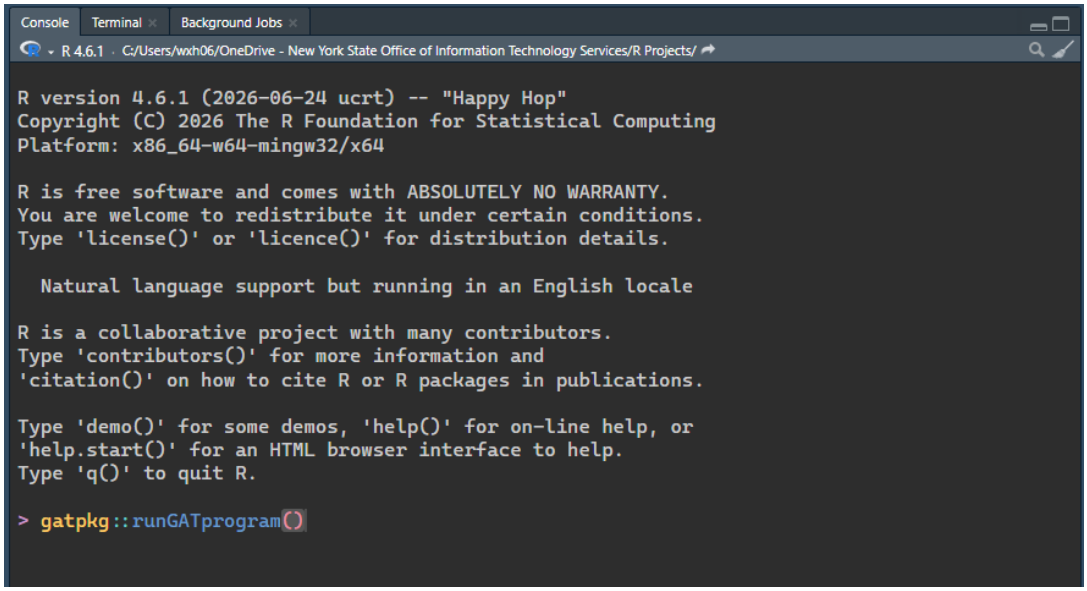
Settings at a glance

Input shapefile	`hftown.shp`
Identifier	`ID`
Boundary	`COUNTY`
Aggregate by	`TOTAL_POP`, minimum 6,000, maximum 15,000
Exclude	`MY_FLAG` equals 1
Merging method	Closest area, by population-weighted centroid
Population file	`hfblockgrp.shp`
Rate	`pop_dens` = 10,000 × `TOTAL_POP` / `AREALAND`
KML	No
Output name	`hftown_agg6K15K_popwt`

Step 0 Launch GAT

In the R console (or RStudio), run:

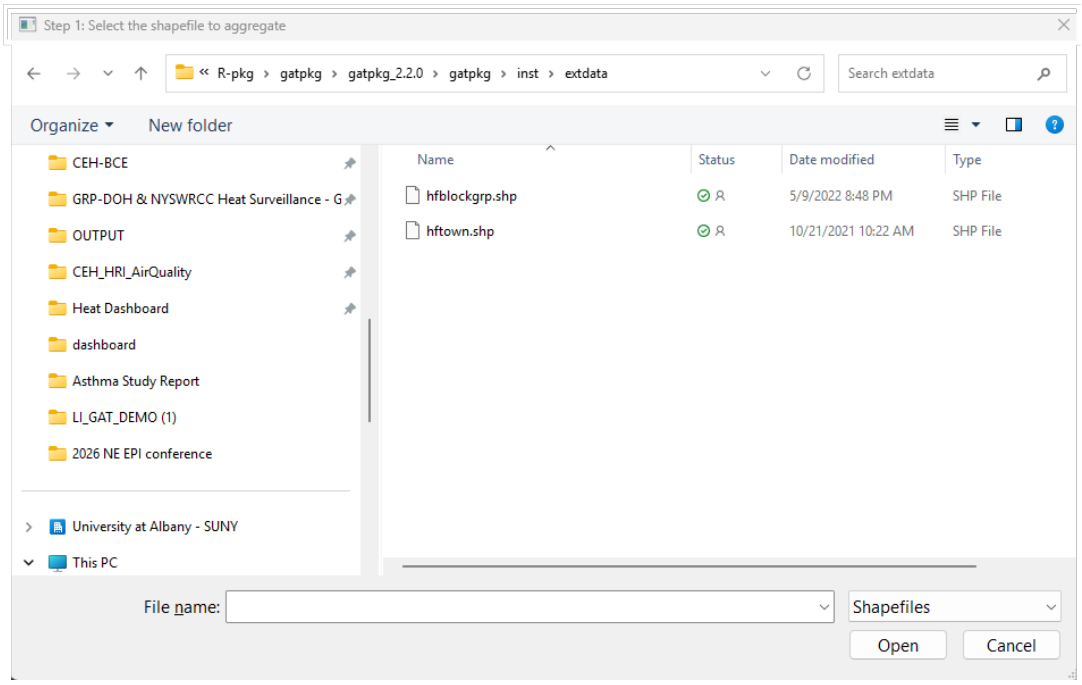
```
gatpkg::runGATprogram()
```



Step 1 Select the shapefile to aggregate

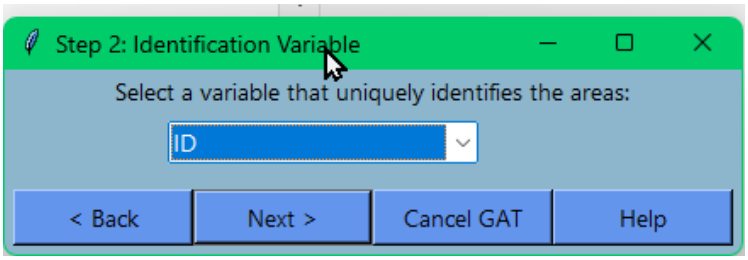
Browse to **hftown.shp** in `gatpkg/inst/extdata or gatpkg/extdata` and click **Open**.

To locate the package directory, you can run find.package("gatpkg") in your R console..



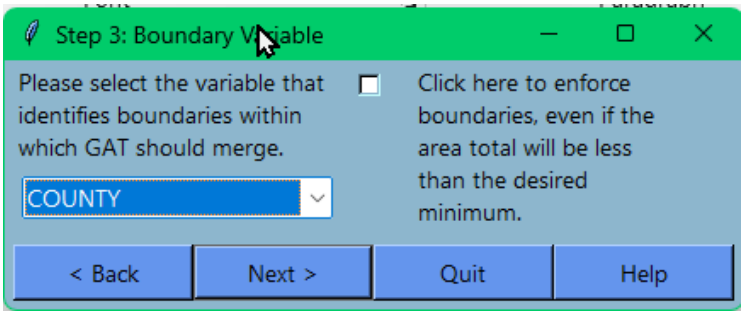
Step 2 Identification variable

Select **ID**, then click **Next >**.



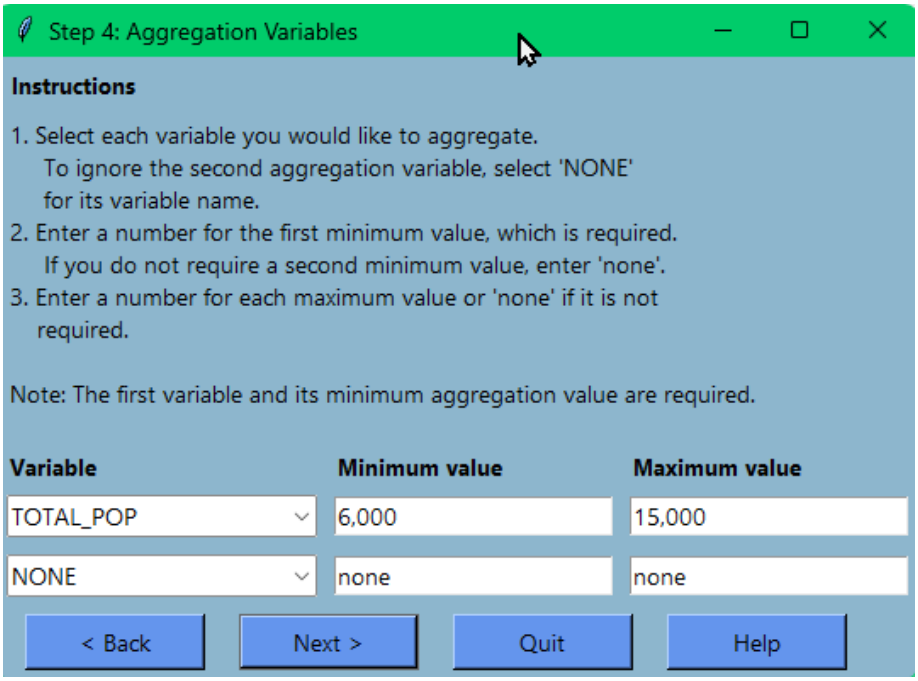
Step 3 Boundary variable

Select **COUNTY**, then click **Next >**.



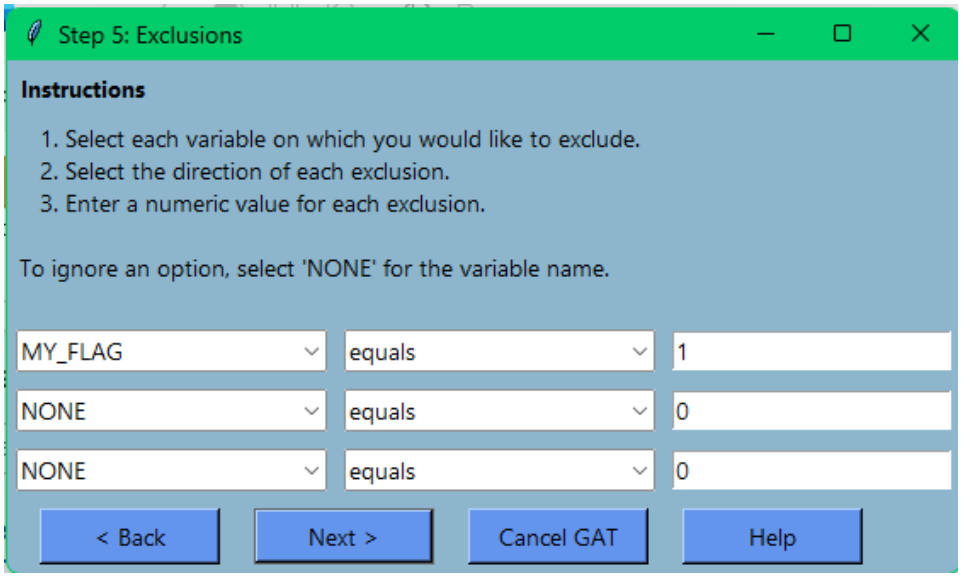
Step 4 Aggregation variables

Select **TOTAL_POP**, enter **6,000** as the minimum value and **15,000** as the maximum value, then click **Next >**.

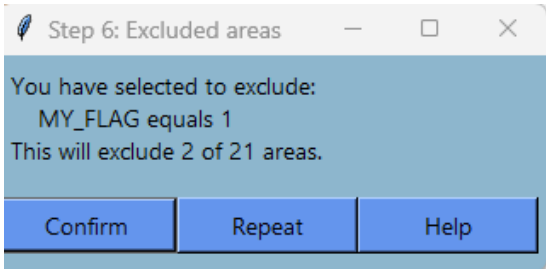


Step 5 Exclusions

In the first row, select **MY_FLAG** and **equals**, then enter **1**. Click **Next >**.

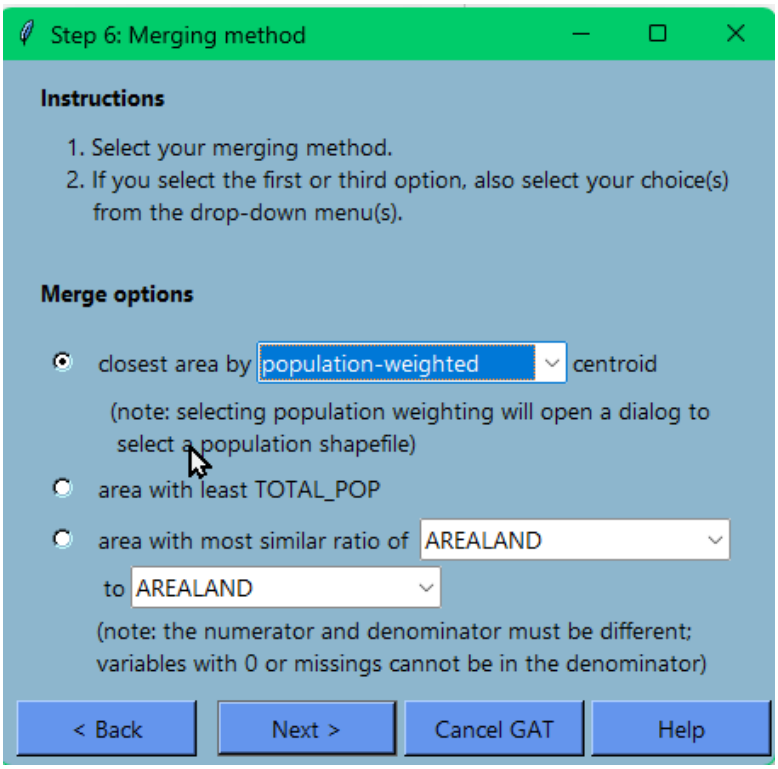


A popup shows how many areas will be excluded. Click **Confirm**.



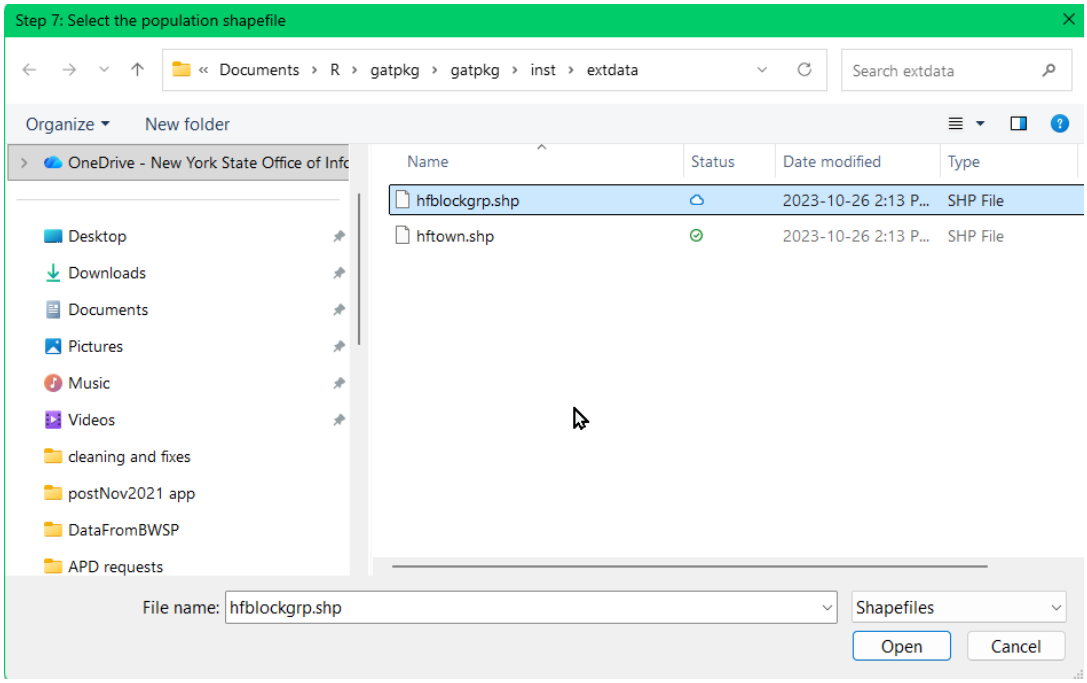
Step 6 Merging method

Select **closest area** by **population-weighted** centroid, then click **Next >**.

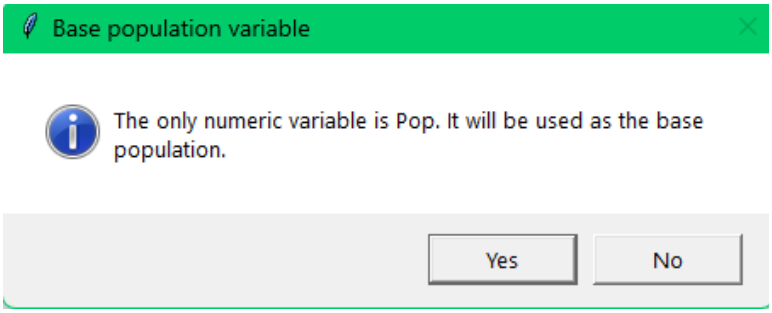


Step 7 Population shapefile

Because you chose population weighting, GAT asks for a population shapefile. Select **hfblockgrp.shp** and click **Open**.

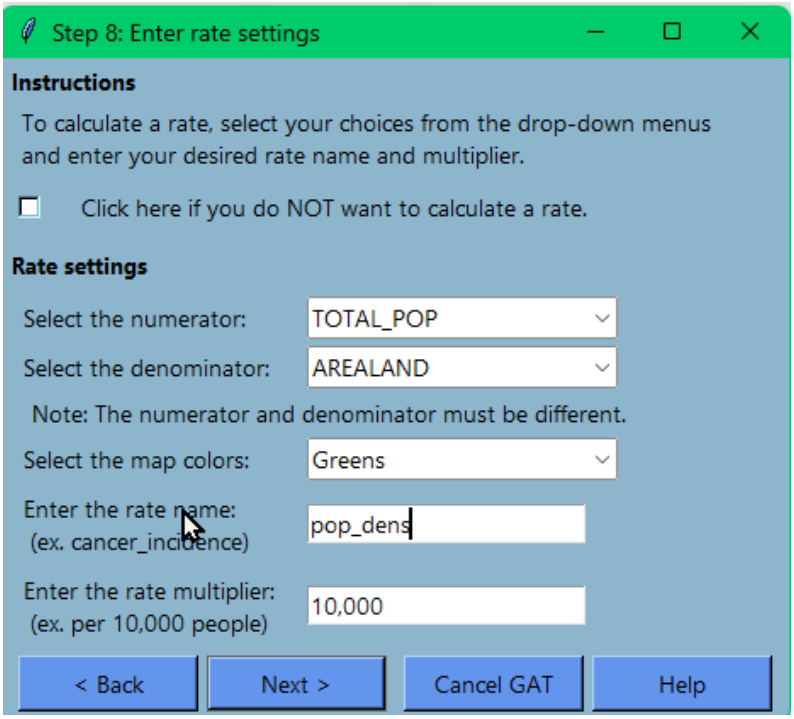


GAT reports that **Pop** is the only numeric variable and will use it as the base population. Click **Yes**.



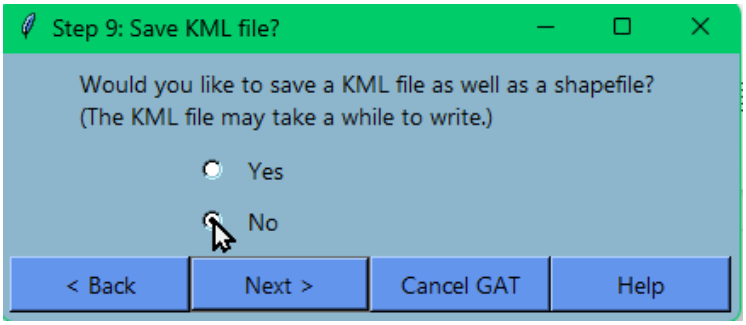
Step 8 Rate settings

Select **TOTAL_POP** as the numerator and **AREALAND** as the denominator, and choose a map color. Enter **pop_dens** as the rate name and **10,000** as the rate multiplier, then click **Next >**.



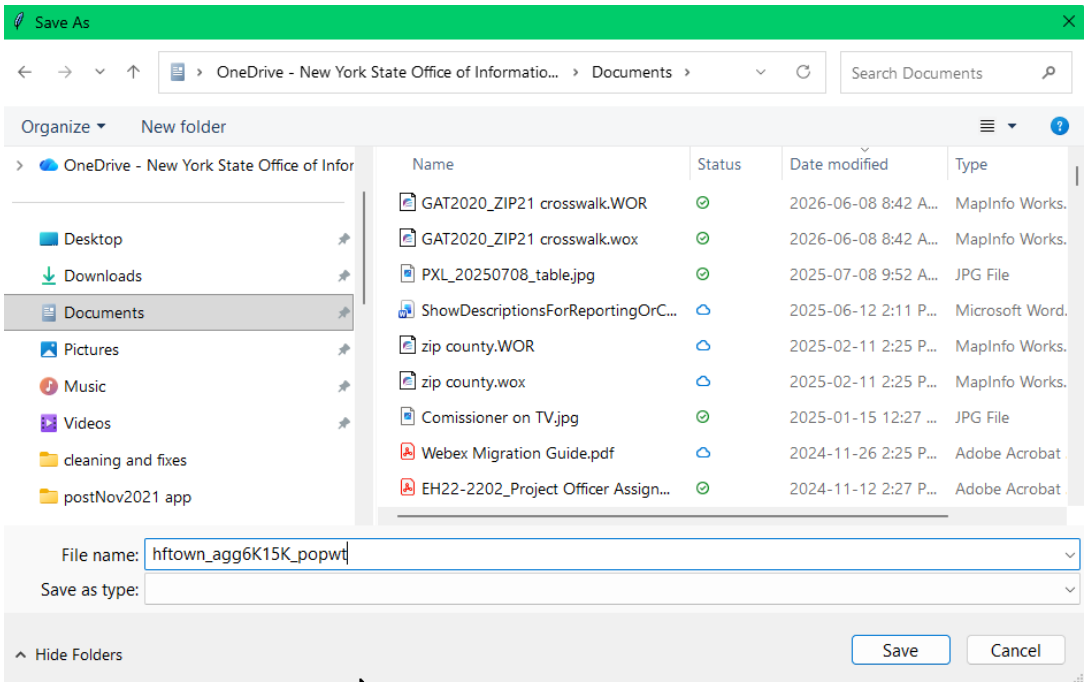
Step 9 Save KML file?

Select **No**, then click **Next >**.



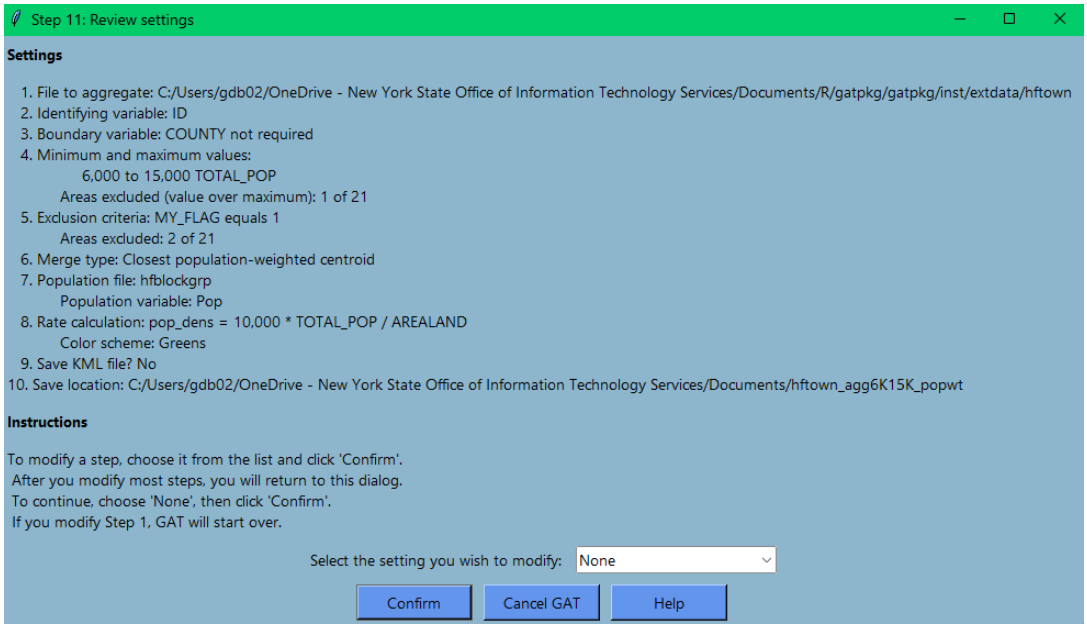
Step 10 Name the output file

In the **Save As** dialog, enter **hftown_agg6K15K_popwt** and click **Save**.



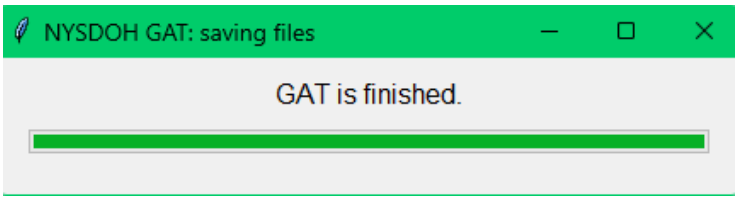
Step 11 Review settings and run

Review the settings. If everything looks right, click **Confirm**. To change a setting, choose it from the drop-down list and click **Confirm**.

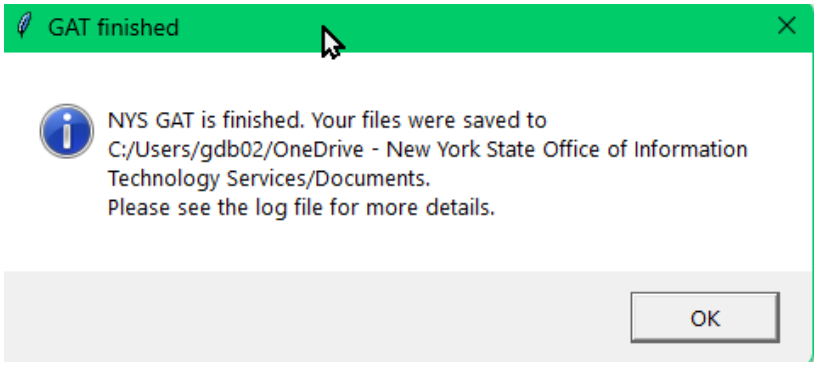


Wait for GAT to finish

A progress box shows GAT's status while it aggregates the areas.



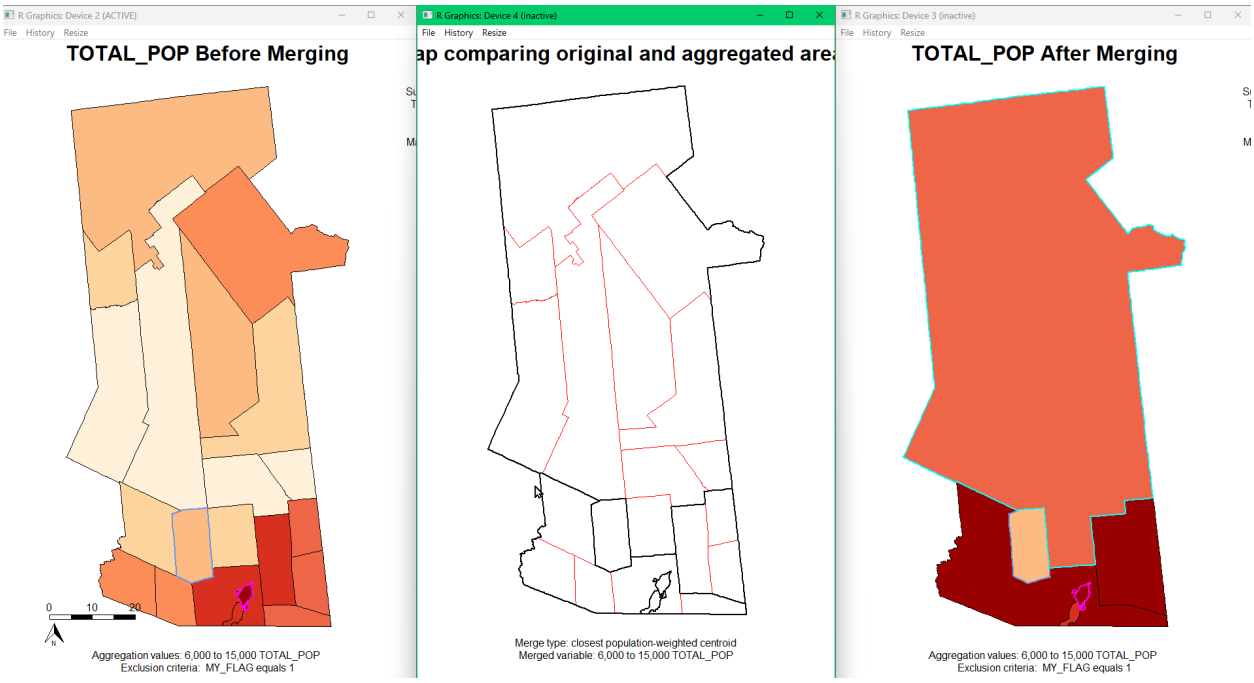
When processing is complete, a message tells you where the files were saved. Click **OK**.



> **Tip:** This sample covers a single county, so it runs quickly. Larger files may take a few minutes or longer.

Check the maps

GAT displays several maps of your aggregated data. Some may be hidden behind others, so rearrange the windows as needed and check for any obvious mistakes.



Output files

GAT saves the following to the location you chose:

- two shapefiles,
- a settings file, and
- a PDF.

Exercise 2: Merging Census Tracts by Most Similar Poverty Ratio

In this exercise you will merge census tracts until each has at least 10,000 residents, excluding unpopulated tracts and tracts where more than half the population lives in group quarters. Tracts are merged with the neighbor whose poverty ratio is most similar.

Before you start

Download and unzip NS_GAT_DEMO.zip from the NYSTracking/gatpkg GitHub repository.

Settings at a glance

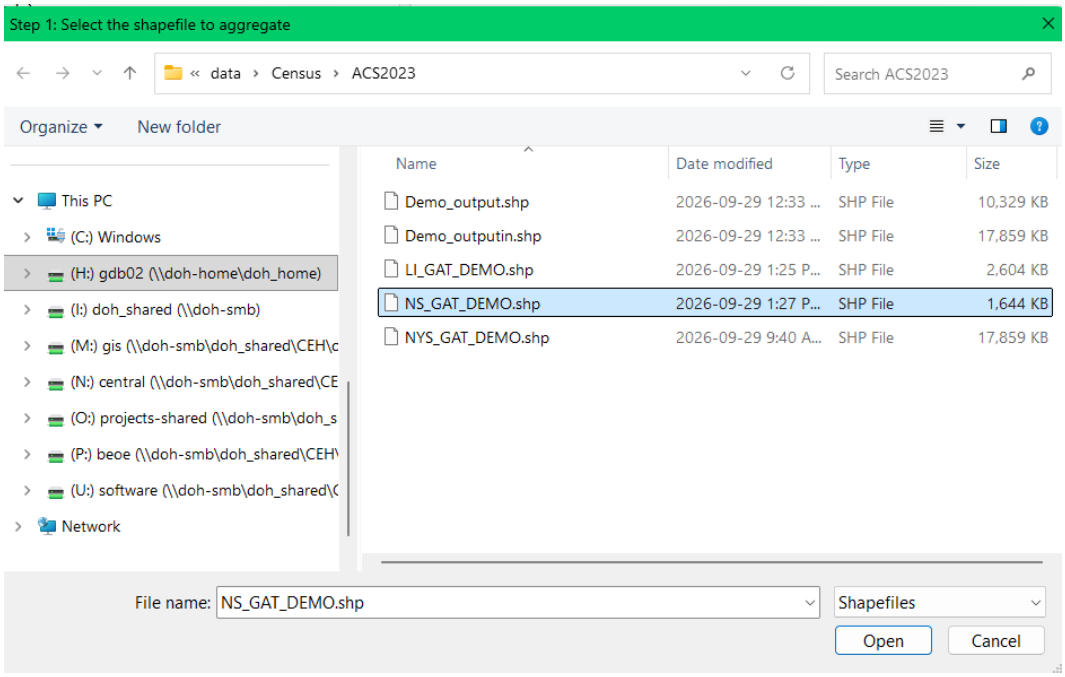
Input shapefile	`NS_GAT_DEMO.shp`
Identifier	`ID` (any of the three options will work)
Boundary	`County`
Aggregate by	`total_pop`, minimum 10,000, no maximum
Exclude	`total_pop` equals 0 `grp_qtr_pc` greater than 0.5
Merging method	Area with most similar ratio of `under_pov` to `pov_determ`
Population file	Not needed (Step 7 is skipped)
Rate	`pov_rate` = 100 × `under_pov` / `pov_determ`
KML	No
Output name	`Demo_output` (for example)

Step 0 Launch GAT

```
gatpkg::runGATprogram()
```

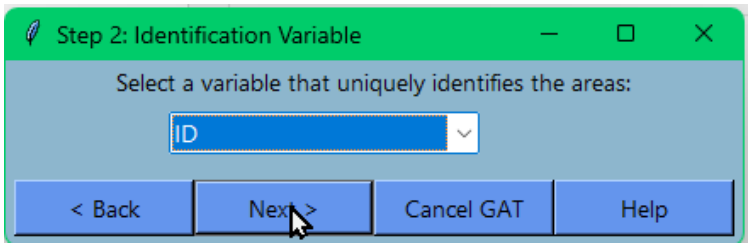
Step 1 Select the shapefile to aggregate

Choose **NS_GAT_DEMO.shp** and click **Open**.



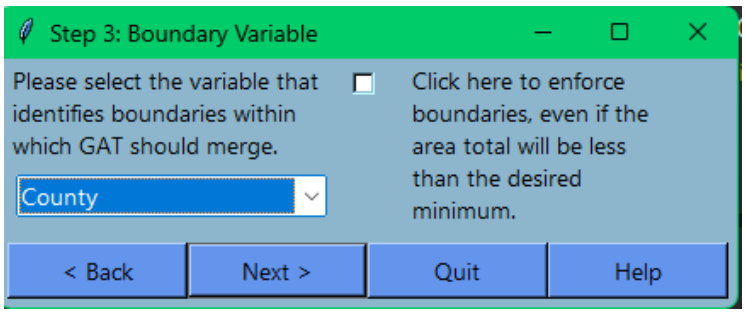
Step 2 Identification variable

Select a field that identifies the tracts (any of the three options will work), then click **Next >**.



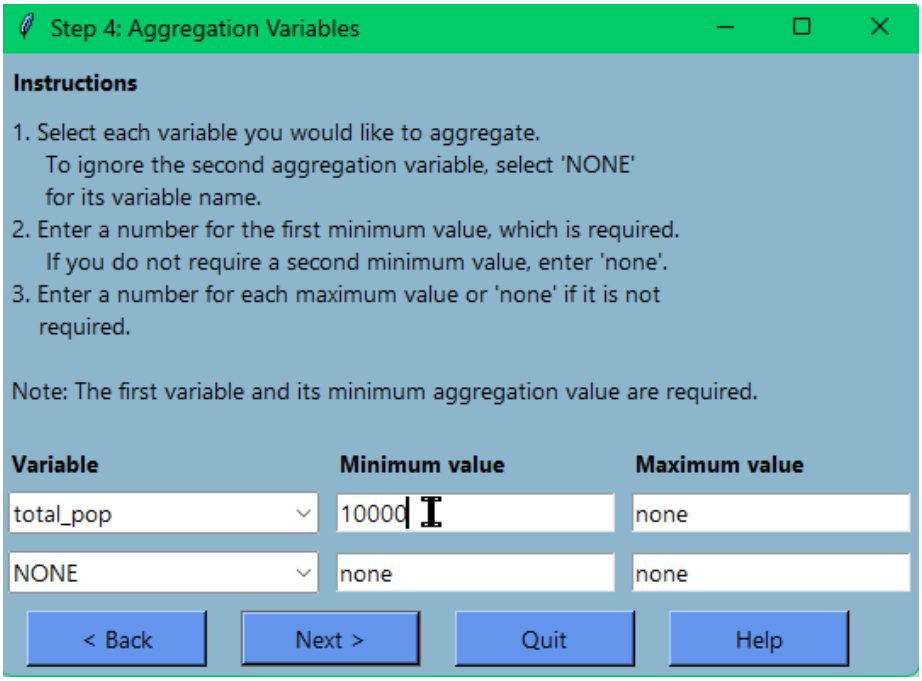
Step 3 Boundary variable

Select **County** and click **Next >**.



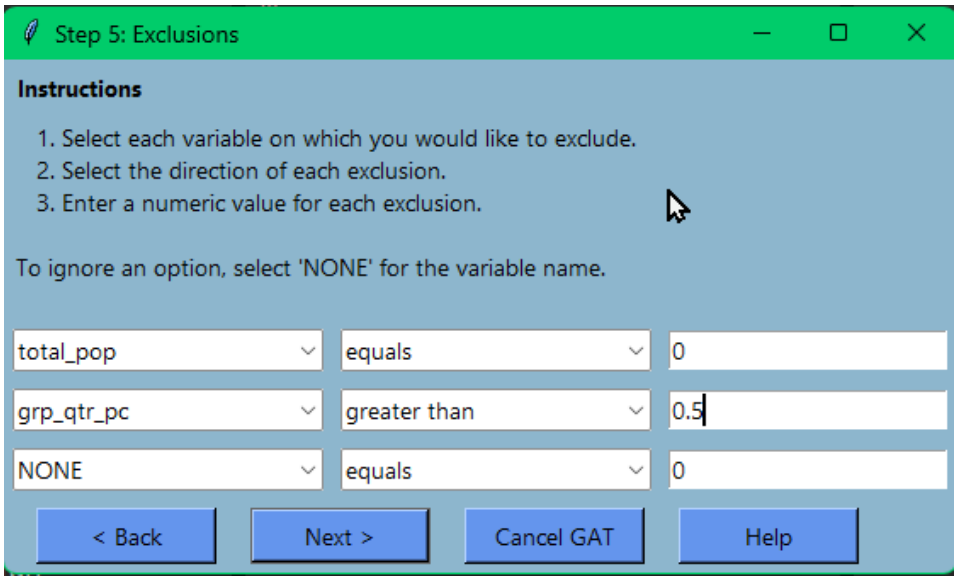
Step 4 Aggregation variables

Set the variable to **total_pop** and the minimum value to **10,000**. Leave the maximum as **none** and click **Next >**.

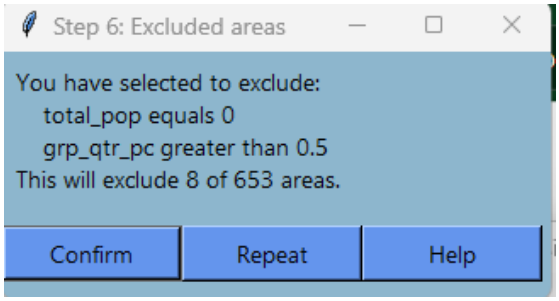


Step 5 Exclusions

In the first row, select **total_pop**, **equals**, **0**. In the second row, select **grp_qtr_pc**, **greater than**, **0.5**. Click **Next >**.

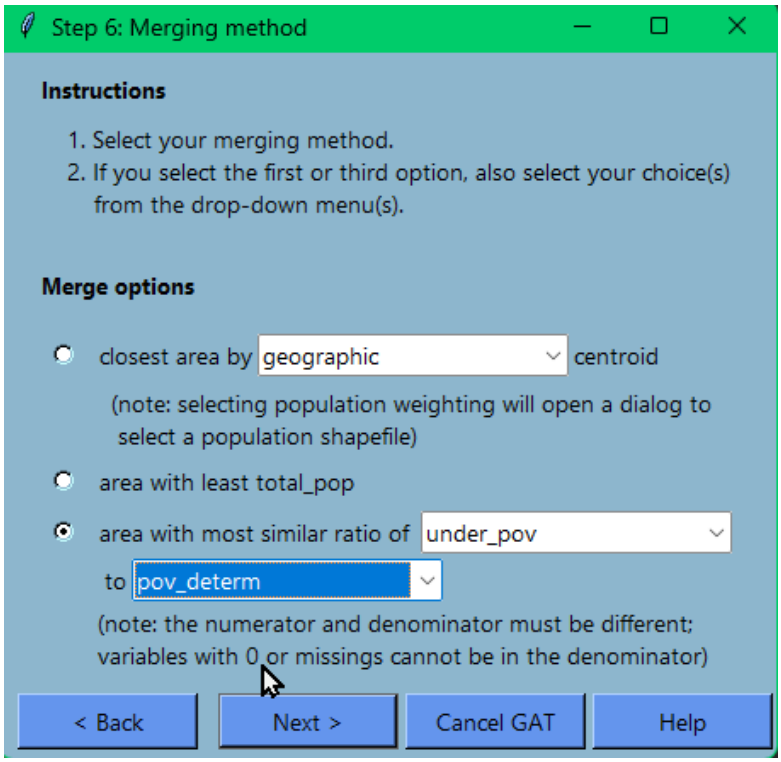


Confirm the exclusions by clicking **Confirm**.



Step 6 Merging method

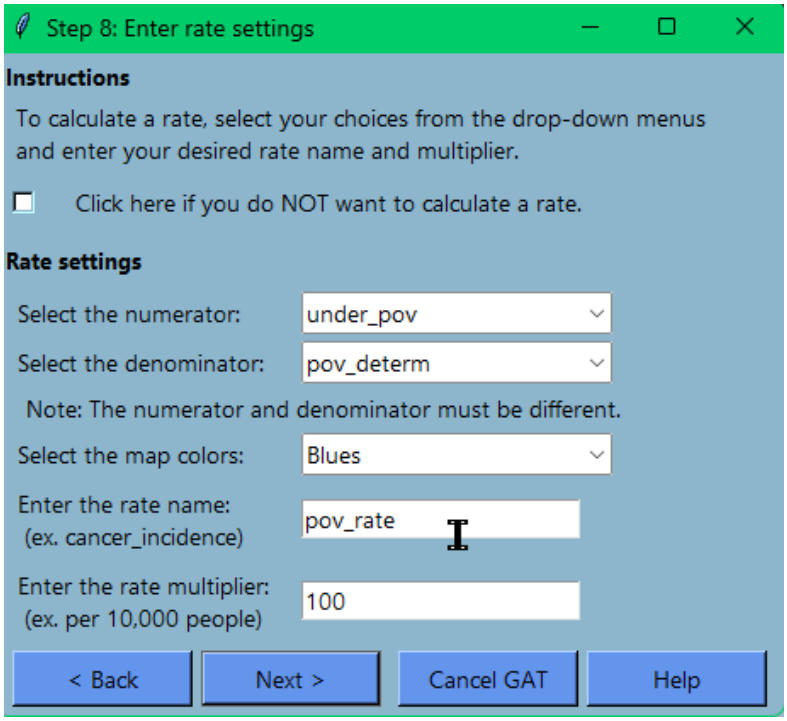
Select **area with most similar ratio of under_pov** to **pov_determ**, then click **Next >**.



> **Note:** Step 7 (population shapefile) appears only when you choose population-weighted merging, so GAT goes straight to Step 8.

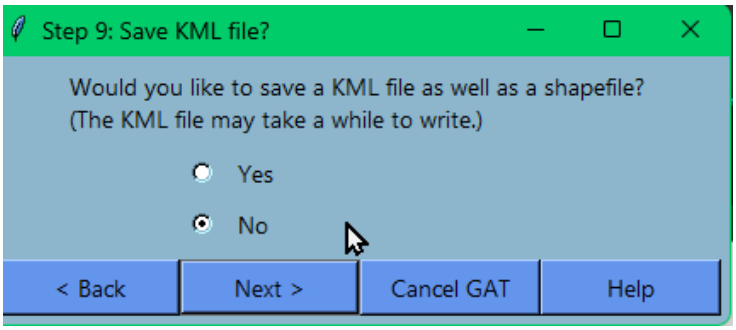
Step 8 Rate settings

Select **under_pov** as the numerator and **pov_determ** as the denominator. Enter **pov_rate** as the rate name and **100** as the rate multiplier, choose a map color, and click **Next >**.



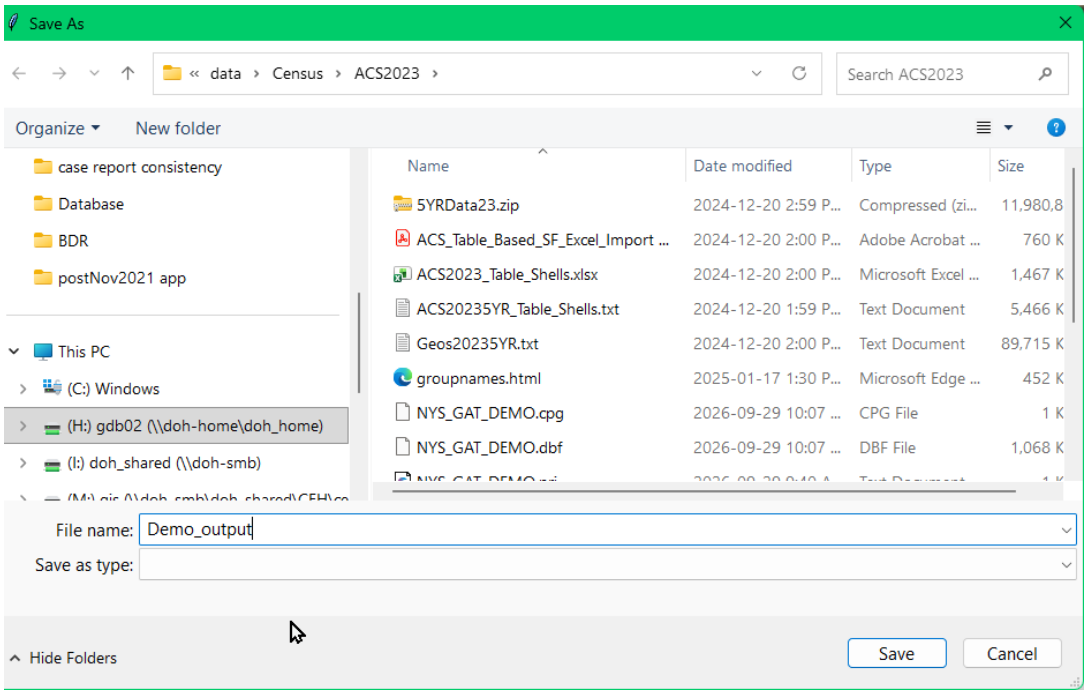
Step 9 Save KML file?

Do not create a KML file. Select **No** and click **Next >**.



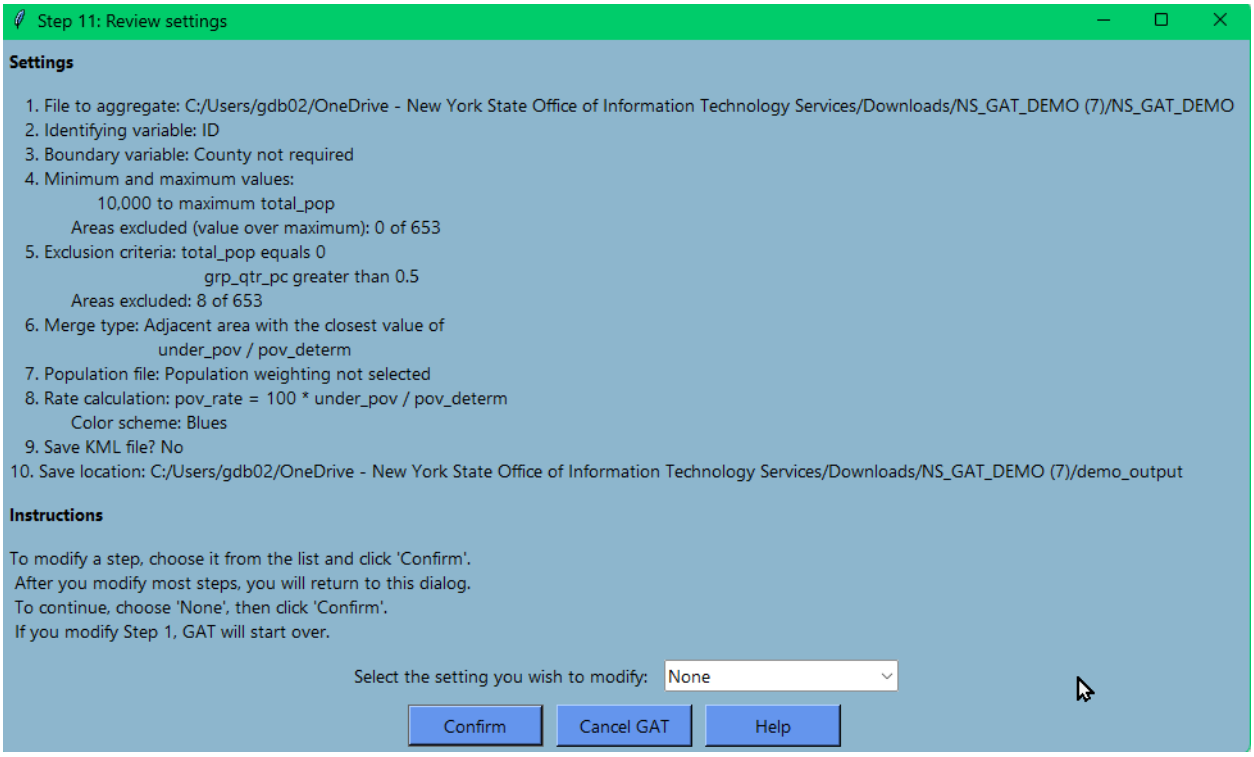
Step 10 Name the output file

Type a file name for the results, for example **Demo_output**, and click **Save**.



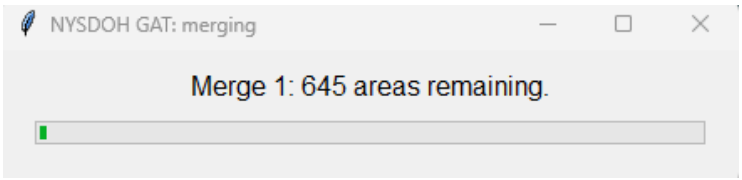
Step 11 Review settings and run

Review the settings and click **Confirm**.



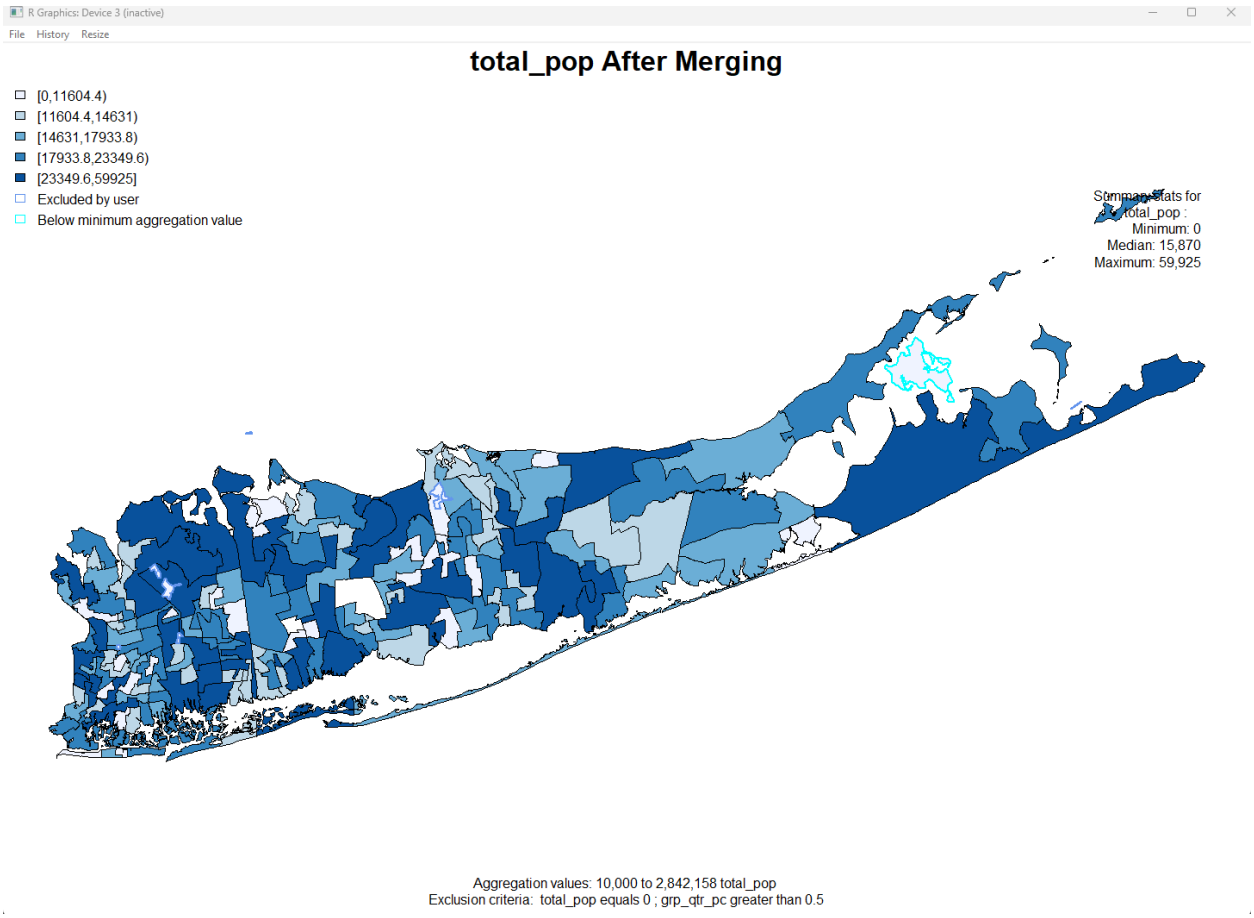
Wait for GAT to finish

This larger file can take around 5 minutes. The progress box shows how many areas remain.



Check the maps

Review the output maps for obvious mistakes.



FYI: The Census Bureau has published an erratum for this area: [Data Collection Error in Wyandanch CDP, New York and Brentwood CDP, New York](#).